

L3 Lead Examiner Report 1901

January 2019

**L3 Qualification in Applied
Science/Forensics and Criminal
Investigations**

**Unit 1: Principles and Applications
of Science I (31617H)**

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Grade Boundaries

What is a grade boundary?

A grade boundary is where we set the level of achievement required to obtain a certain grade for the externally assessed unit. We set grade boundaries for each grade, at Distinction, Merit and Pass.

Setting grade boundaries

When we set grade boundaries, we look at the performance of every learner who took the external assessment. When we can see the full picture of performance, our experts are then able to decide where best to place the grade boundaries – this means that they decide what the lowest possible mark is for a particular grade.

When our experts set the grade boundaries, they make sure that learners receive grades which reflect their ability. Awarding grade boundaries is conducted to ensure learners achieve the grade they deserve to achieve, irrespective of variation in the external assessment.

Variations in external assessments

Each external assessment we set asks different questions and may assess different parts of the unit content outlined in the specification. It would be unfair to learners if we set the same grade boundaries for each assessment, because then it would not take accessibility into account.

Grade boundaries for this, and all other papers, are on the website via this link:

<http://qualifications.pearson.com/en/support/support-topics/results-certification/grade-boundaries.html>

Unit 1: Principles and Applications of Science I

Grade	Unclassified	Level 3			
		N	P	M	D
Boundary Mark	0	10	21	39	57

Individual Questions

Biology

Q1ai

The majority of learners were able to correctly identify the golgi apparatus from the figure.

This response was awarded 1 mark.

(a) (i) Name organelle X in Figure 1.

(1)

Golgi aparatus

There were quite a few students who attempted to make an educated guess at endoplasmic reticulum.

This response was awarded 0 marks.

(a) (i) Name organelle X in Figure 1.

(1)

Smooth Endoplasmic reticulum

Q1a_{ii}

The question was mostly well answered

This response was awarded 1 mark.

(ii) Identify organelle Y in Figure 1.

- ☒ **A** centriole
- ☒ **B** endoplasmic reticulum
- ☒ **C** mitochondrion
- ☒ **D** vesicle

Q1aiii

This question was also mostly well answered.

This response was awarded 1 mark.

(iii) Identify the function of the nucleolus.

- ☒ **A** forms spindle fibres during cell division
- ☒ **B** makes RNA and ribosomes
- ☒ **C** regulates cellular activity
- ☒ **D** synthesises and transports lipids and carbohydrates

A common wrong answer was C, the learners' likely mistaking nucleolus for nucleus.

This response was awarded 0 marks.

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Q1bi

Overall this question was well answered. The most common responses included chloroplast, cell wall and vacuole, although a few mentioned tonoplast, plasmodesmata and amyloplast. The learners rarely qualified the vacuole in the plant cell as large or permanent which would have shown a deeper understanding.

This response was awarded 2 marks.

(i) Name **two** structural features that are found only in plant leaf cells and not in animal cells.

(2)

- 1 plasmadomata
- 2 chloroplast

Many candidates put chlorophyll, indicating a lack of understanding of the chemical or a confusion with chloroplast. Some candidates also commented on differences between plants tissue rather than plant and animal cells, for example, presence of phloem and spongy mesophyll

This response was awarded 1 mark.

(2)

- 1 Chloroplasts
- 2 palisade cells

Q1bii

This question was poorly answered, with fewer than half of the responses being credit-worthy. Centrioles was a common response with scoring learners, although spelling was generally poor. Very few learners stated cilia.

This response was awarded 1 mark.

Centrioles

A common mistake was misremembering centriole as centricle.

This response was 0 awarded marks.

centricles

A wide variety of incorrect answers were given for this question including nucleus, cell membrane, ribosomes, mitochondria, golgi apparatus and endoplasmic reticulum.

Some learners stated organelles present in plant cells and not in animal cells, thus answering the previous question again. Some learners confused prokaryotes and eukaryotes stating plasmids or nucleoid in response to this question.

Q2a

Most learners were awarded a mark for correctly completing the first missing word from the definition by providing an acceptable synonym for specific, commonly giving unique, different, and particular. Few learners were awarded the second mark for "differentiation".

This response was awarded 2 marks

A cell becomes specialised when its structure is altered. This enables a cell to have a specific function. This process is called cellular differentiation.

Those learners who failed to achieve the first marking mostly did so by simply repeating 'special' from the stem of the question. This was also seen in the majority of responses for the second marking point with most learners giving 'specialisation' and division.

This response was awarded 1 mark

A cell becomes specialised when its structure is altered. This enables a cell to have a ~~specialised~~ unique function. This process is called cellular division.

Q2bi

The majority of learners were able to correctly identify the correct answer.

This response was awarded 1 mark.

(i) Identify the type of cell labelled W in Figure 3.

- ☒ A ciliated
- ☒ B columnar
- ☒ C endothelial
- ☒ D stratified

Q2bii

There were a good number of learners who answered with the idea of a multi-lobed nucleus giving flexibility to squeeze through the pores in the capillary. The subject was quite well understood by the learners on the whole. The idea of flexibility was widely seen even if the word 'flexible' was rarely used. Many learners described the idea of the shape being able to change, which was accepted.

This response was awarded 2 marks.

A neutrophil has a multi-lobed nuclei which allows the white blood cell to squeeze through small gaps to reach the site of infection or injury faster.

Some learners confused neutrophils with other specialised cells from the specification content and explained how the structure of red blood cells or sperm allow them to perform their function. Some learners stated that neutrophils contain a large number of lysosomes and, whilst this is correct, does not answer the question as it is not an adaptation which allows them to leave capillaries.

Some responses lost marks by stating that the neutrophils could move through the capillaries, therefore repeating the question.

Quite a few responses confused the idea of the cell being flexible with the cell actually changing size, saying that it got smaller to squeeze through the capillaries then got bigger again. There were also a significant number of responses in which the learner had confused neutrophils with other organelles, describing synthesis and transportation of lipids, proteins or carbohydrates and some describing energy production. There were also a number of responses that vaguely referred to immune response.

This response was awarded 0 marks

The structure allows it to exit the cell & engulf
bacteria outside, it's easy maneuvering allowing
it to move easily.

Q2biii

This question was generally well-answered and the subject was quite well understood, with many of the learners achieving marking point 3 and a significant number earning two marks for the concept of 'destroying pathogens'. Some even gave detail about the phagocytic vesicle fusing with the lysosome. Most referred to enzymes in the lysosome. Very few learners mentioned the formation of phagolysosome.

This response was awarded 3 marks.

Lysosomes carry an enzyme for digesting pathogens. This is key for a neutrophil because they are a type of white blood cell so their function is to destroy pathogens so the function of lysosomes allow them to do this.

Some learners showed confusion over what a lysosome is, describing them as enzymes or organelles within the cell and made general statements about 'getting rid' of pathogens or 'breaking down' waste material. Cell division was a common theme among the incorrect responses and there were a few that mentioned digestive enzymes.

Many incorrect responses described the lysosome engulfing the pathogen and a significant number mentioned protein synthesis. There was even the occasional reference to lysosomes as an energy source.

This response was awarded 1 mark

Neutrophils contain lysosomes. Lysosomes are a main part of a neutrophil, it allows them to attack a pathogen or unwanted foreign body to protect the individual from infection.

Q2c

The full range of marks were observed in this question, but very few learners achieved all three.

This response was awarded 3 marks. The learner has given 1643 on the answer line, which is an acceptable rounding of the correct evaluation.

$$\begin{aligned}
 & 3.5 \times 10^9 = 100\% \\
 & (6.1 \times 10^{10}) - (3.5 \times 10^9) = \text{increased by } (3) \text{ } 5.75 \times 10^{10} \\
 & \frac{5.75 \times 10^{10}}{3.5 \times 10^9} \times 100 \\
 & = 1643 \\
 & \text{percentage increase in the neutrophil count} = 1643\%
 \end{aligned}$$

Marking point 1 was by far the most common mark achieved, most using the third method to calculate their answer. Very few learners attempted the second method. A lot of candidates took the numbers out of standard form to do the calculations. A few learners achieved 1 or 2 marks by using the third method from the mark scheme but few took away 100 to give the correct response.

Many learners appeared to have difficulty with the numbers being in standard form, which often led them to get the difference wrong. A significant number of learners got their divisions upside down and a number don't appear to have understood that it is possible to have an increase of more than 100%.

This response was awarded 2 marks.

$$\begin{array}{l}
 3.5 \times 10^9 = 3500\,000\,000 \\
 6.1 \times 10^{10} = 61\,000\,000\,000 \\
 \hline
 \frac{6.1 \times 10^{10} - 3.5 \times 10^9}{6.1 \times 10^{10} + 3.5 \times 10^9} = 0.49 \times 100 \\
 = 49
 \end{array}$$

percentage increase in the neutrophil count = 49 %

In this example, the learner has used the first method from the mark scheme. They have substituted the values to calculate the difference in their working out and is shown on the top of their division. Therefore, Mp1 can be awarded. The learner has not divided their calculated difference by the original value and so therefore cannot be awarded with mp2. However, the learner has then multiplied their division by 100 and so can be awarded mp3 as an error carried forward mark.

Q3a

This question was well answered, but with many learners answering phonetically or misspelling Ranvier.

This response was awarded 1 mark.

Nodes of ranvier

Many learners restated myelin sheath or incorrectly thought that the region was the synapse or axon although some just said node, which was insufficient for the mark.

This response was awarded 0 marks.

(1)

myelin sheath node of radion

Q3b

Overall, there were many very good responses with a significant number of learners achieving 4 marks. The learners showed a good understanding of the myelin sheath being wrapped around the axon and/or acting as an insulator and many included Schwann cells and lipids/fat. However, very few learners gave "has a nucleus" as a response.

This response was awarded 3 marks

Myelin sheath wraps around the axon.
They contain many layers of myelin sheath.
inside the myelin sheath is a Schwann
cell which controls the myelin sheath.
They are found on the outer layer of the sheath.
They are an insulating layer for the axon
and between sheaths are nodes of Ranvier.

Most learners who lost marks did so by focusing on the function of the sheath rather than its structure. A lot of answers merely reiterated the stem, were too vague.

This response was awarded 1 mark for Schwann cell.

Myelin sheath are white.
It contains a schwann cell that which
make myelin sheath,
increases electrical impulses for
speed.

Q3c

This question was quite poorly answered with very few answers mentioning anything beyond insulation and saltatory conduction (commonly referred to as 'impulse jumping'). Very rarely were references to loss of ions/less shielding seen.

This response was awarded 2 marks

(2)

this means that electrical impulses can not bounce through the nodes to speed up. lack of insulation means that electrical impulses will be slower

Many errors were seen in the responses to this question. Most commonly a confusion between insulation and conduction or simple descriptions of the diagram e.g. 'myelin is broken/damaged' were given. A lot of learners simply repeated the stem of the question. Of the marking points available, few learners achieved points three to six, often failing to apply their knowledge and relate it to a relevant marking point, especially in reference to action potentials. Some learners showed confusion over speed and mitochondria and the need for energy.

This response was awarded 0 marks

(2)

This is because chemical reactions work at a slower rate, this therefore means the chemical information can't travel through the neurons as quickly. As the structure is damaged.

Q4

This question was, in general, quite poorly answered. Most learner responses described the formation of atherosclerosis but failed to mention the contribution of smoking, restricting responses to level 2, 3-4 marks. Many low scoring responses described damage to cilia and the formation of mucus. Some learners scored marks for including the migration of white blood cells to the damaged area and a very small minority showed an understanding of foam cells.

Quite a lot of responses were able to link smoking to the damage to cells and the resultant inflammation followed by a build-up of plaque under the cells. Some linked carbon monoxide from tobacco smoke (although some confused it with carbon dioxide) to reduced oxygen-carrying capacity and increased blood pressure/strain on heart and linked nicotine to increased stickiness of platelets and formation of thrombus.

This response was awarded 5 marks.

Discuss how smoking is a risk factor for the development of atherosclerosis.

(6)

When a human smokes, those particles or bad cholesterol (LDL) cholesterol, deposits on the endothelial tissue, when it sits on the tissue, white blood cells come to engulf the particles or the bad cholesterol, which ends up forming foam cells. The accumulation of these foam cells due to more smoking and other activities, leads to a fatty plaque. Calcium ions also accumulate and deposit on the foam cells which leads to atheroma. Deposition of more atheroma which is white blood cells, foam cells and calcium ions leads to a fatty plaque thus

causing the development of atherosclerosis. The human artery is scarred, enlarged and it blocks the efficiency in the flow of blood which could lead to blood clots or heart cardiac disease and high blood pressure. This happens when the blood is struggling to by pass the fatty plaque which may lead to reduced blood flow. Blood cannot flow smoothly in the artery if its affected by atherosclerosis.

Many learners were not able to distinguish between the effects of smoke on lungs and arteries, with a significant number discussing how hot smoke could burn cilia. A significant number also suggested that it was tar/mucus that blocked the arteries and some who were under the impression that plaques were formed of food and saliva. There was a lot of repetition in some answers

Overall, there was a lot of confusion, mainly to do with damage to the lungs (cilia damage, build-up of mucus, alveoli damage etc.) although this could have been creditworthy had they linked it to lower oxygen levels in the blood leading to higher blood pressure. Responses often contained discussions of problems caused by atherosclerosis rather than why smoking is a risk factor.

This response was awarded 2 marks.

Firstly smoking is a risk factor of causing atherosclerosis because in the bodies arteries there will be an increase of blood pressure. This has a negative affect on the bodies ^{arteries} ~~arteries~~, this therefore means mucus is trapped in the artery and the amount of blood flow is reduced causing muscles to have a less volume of oxygenated blood.

Smoking is also a risk factor for causing atherosclerosis, this is because the heart is strained and put under strain, this means it has to work harder to pump blood, in addition the smaller area for the blood to travel through means the arteries have a higher pressure within them.

Furthermore in gaseous exchange blood won't be able to do that as effectively because the arteries can't carry as much blood, this therefore means red blood cells have a decreased store of haemoglobin.

Smoking also a risk factor for producing atherosclerosis because the increase store of mucus and mucus production as the body constantly tries to clear it.